INCH-POUND
MIL-R-28750/10B
14 November 1988
SUPERSEDING
MIL-R-28750/10A(EC)
3 November 1982

MILITARY SPECIFICATION SHEET

RELAYS, SOLID STATE, SEALED, OPTICALLY ISOLATED, ZERO VOLTAGE TURN ON, 25 AMPERES, 250 VOLTS MAXIMUM, 45-440 HZ, POWER SWITCHING, SPST (N.O.)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the relays described herein shall consist of this specification and the latest issue of MIL-R-28750.

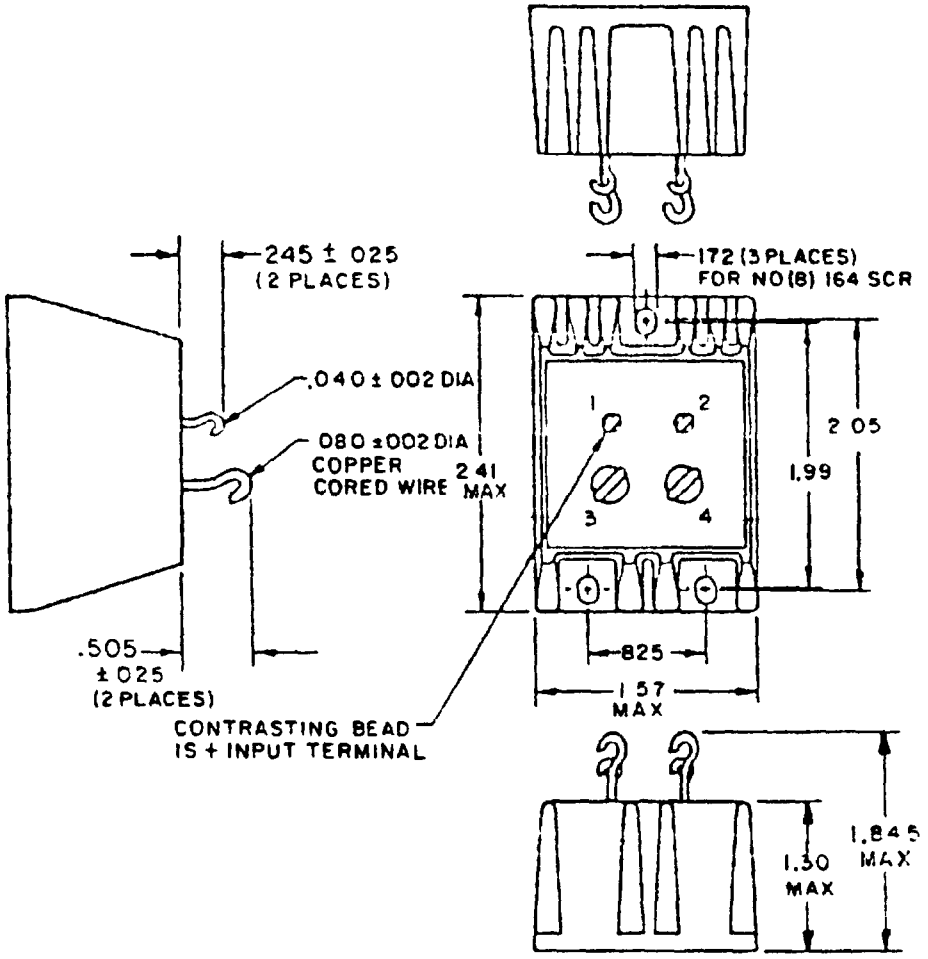
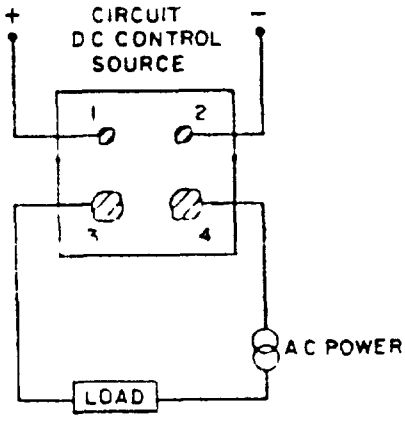


FIGURE 1. Outline drawing and dimensions.

(B) denotes change

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CIRCUIT DIAGRAM
TERMINAL VIEW

Inches	mm
.002	.05
.025	.63
.040	1.02
.080	2.03
.172	4.37
.245	6.22
.505	12.83
.785	19.94
1.30	33.0
1.57	39.9
1.645	46.86
1.99	50.5
2.05	52.1
2.41	61.2

NOTES
1. Dimensions are in inches
2. Metric equivalents are given for general information only
3. Unless otherwise specified, tolerance is ± .010 (.25 mm)
4. Terminal numbers shown above are for reference only, and do not appear on the header.

FIGURE 1. Outline drawing and dimensions - Continued

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REQUIREMENTS

MECHANICAL DATA

Weight 6 ounces maximum.

Seal (Solder or weld) MIL-STD-202, method 112, test condition A, except temperature +125°C +0°C, -5°C.

Dimensions and configuration: See figure 1

Terminals

Terminal strength: 5 pounds pull

Terminal solderability: Applicable.

INPUT DATA

Input voltage range 4.0 to 32.0 V dc.

(B) Assured turn-on voltage: $\leq$ 4.0 V dc.

(B) Assured turn-off voltage $\geq$ 1.0 V dc

Input current 16 mA maximum at 32 V dc

Turn-on time $\frac{1}{2 \text{ (line frequency)}}$ maximum

Turn-off time $\frac{1}{\text{(line frequency)}}$ maximum.

Bias current Not applicable.

OUTPUT DATA

Output voltage range 25 to 250 V ac, 45 to 440 Hz

Rated output current 25 A, ac maximum (see figures 2 and 3).

(B) Rated output voltage 220 V maximum, 45 to 440 Hz

Output voltage drop 1.5 V rms maximum.

(B) Output leakage current. 10 mA, ac maximum at 220 V ac, 400 Hz.

Crosstalk: Not applicable

(B) Transient voltage. MIL-STD-704A, figure 3, curve 1, 180 volts, 5 seconds

Electric system spike Not applicable

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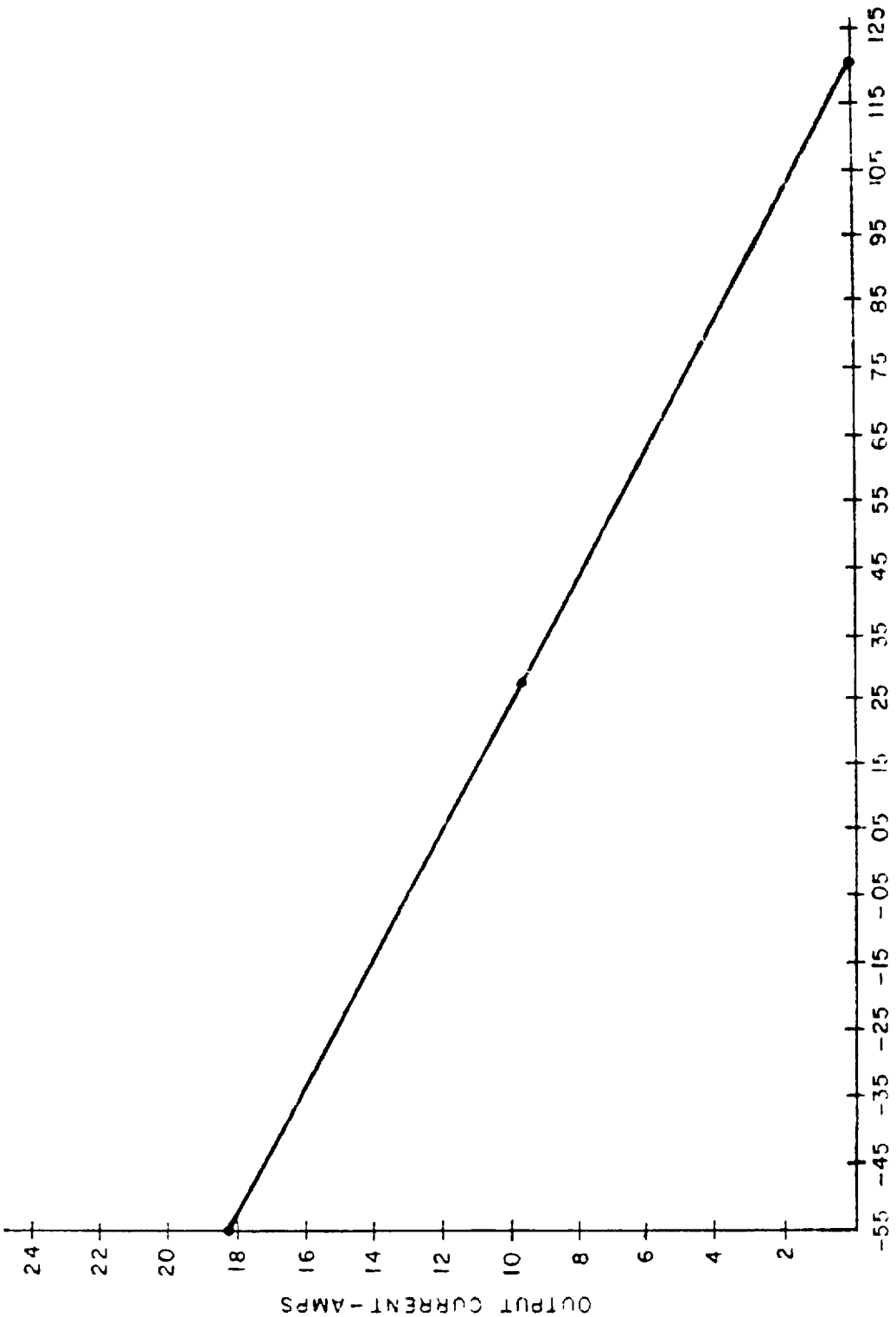


FIGURE 2. Output current versus ambient temperature.

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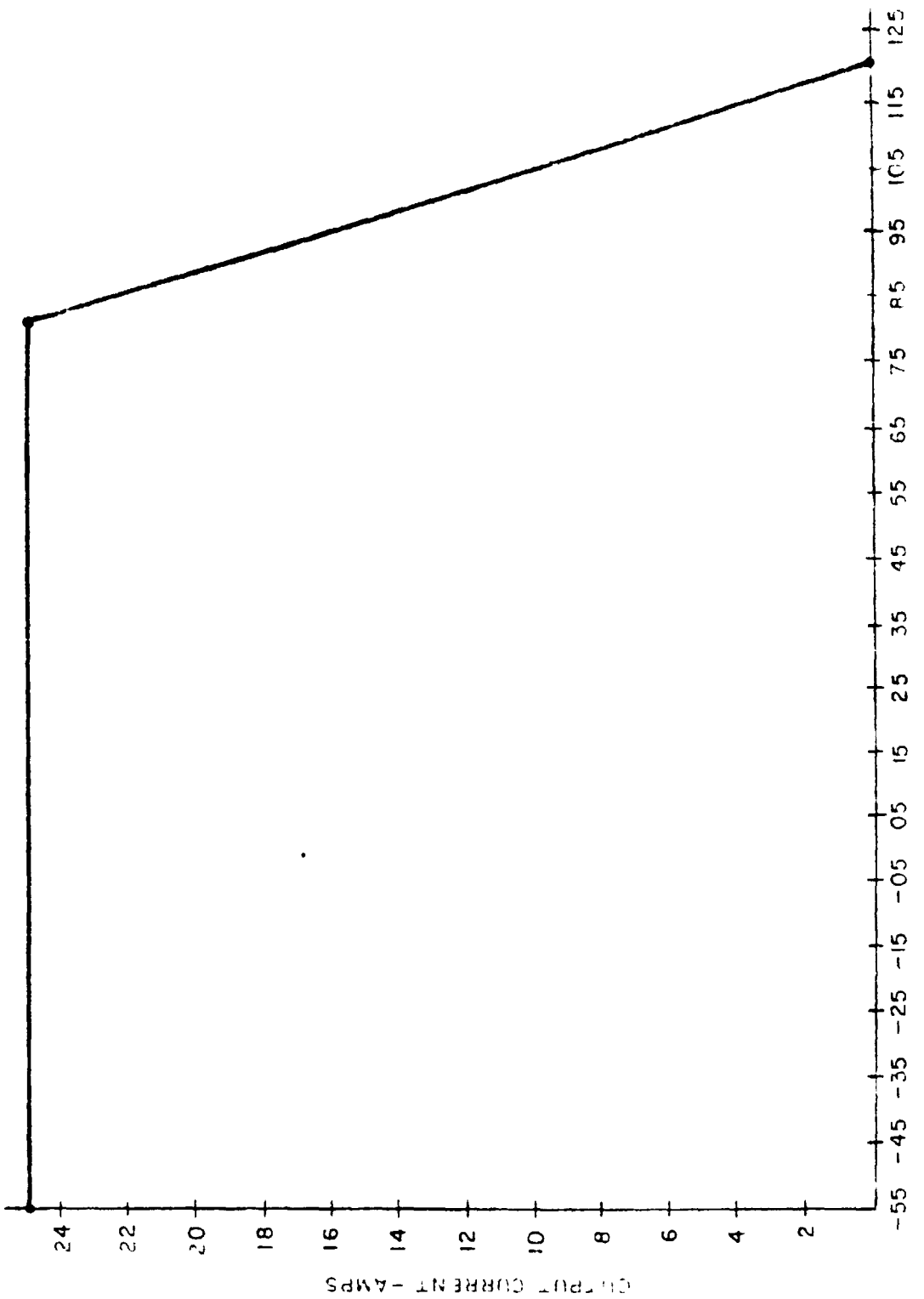


FIGURE 3. Output current versus case temperature.

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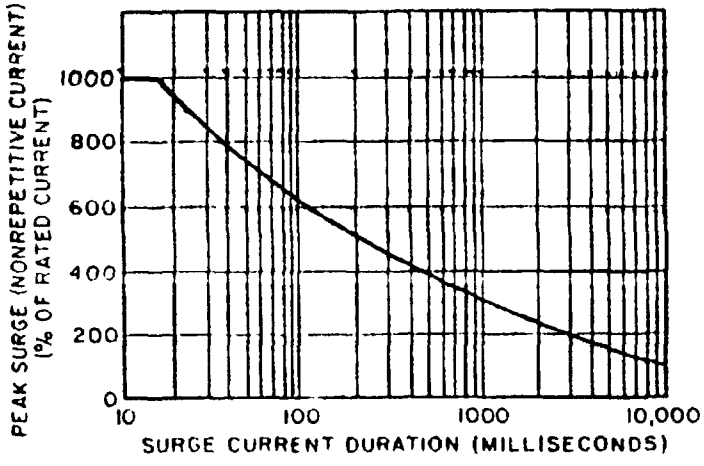


FIGURE 4. Peak surge current vs. surge current duration.

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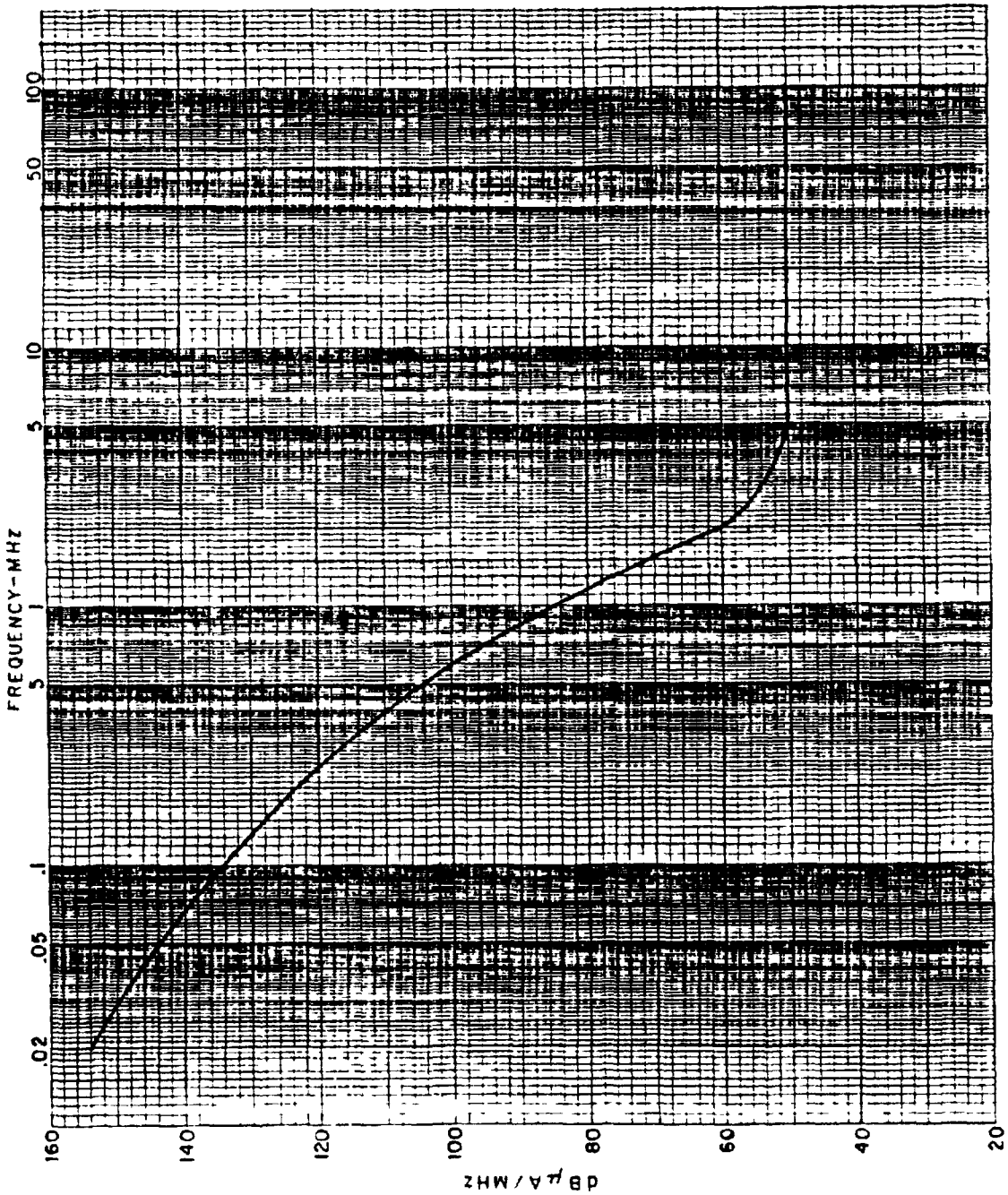


FIGURE 5. Broadband conducted emissions.

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Ⓑ Overload 80 amperes.

DC offset voltage: ±150 mV maximum

Waveform distortion 4 V rms maximum from 10 percent to 100 percent rated output current

Ⓑ Zero crossover voltage (at +25°C). M28750/10-001 = 0.0 ±15 volts peak maximum.
M28750/10-002 = 0.0 ±40 volts peak maximum.

ELECTRICAL DATA.

Dielectric withstanding voltage 1,500 V ac maximum at 60 Hz.

Insulation resistance: 100 megohms minimum.

Isolation. 20 picofarads maximum.

Power dissipation: 38 watts maximum.

ENVIRONMENTAL DATA

Temperature:

Operation. -55°C to +110°C.

Storage: -55°C to +125°C.

Ⓑ Shock (specified pulse) MIL-STD-202, method 213, test condition F (1500 g's).

Ⓑ Vibration 30 g's, 10 to 3,000 Hz

Moisture resistance: Not applicable.

Electromagnetic interference: Maximum broadband conducted emission on power lines with 115 V ac, 400 Hz, 10 amperes resistive load during steady state and switching conditions (see figure 5).

Resistance to soldering heat: Applicable.

Salt spray (corrosion): In accordance with method 1041 of MIL-STD-750.

Ⓑ Part or identifying number (PIN): M28750/10-001, or M28750/10-002, W, Y, or X to indicate the applicable screening level as specified in MIL-R-28750.

Qualification: To qualify either M28750/10-001 or -002 all qualification inspections of MIL-R-28750 shall be performed. In order to qualify M28750/10-001 and -002 submit M28750/10-001 to qualification inspection of MIL-R-28750 as applicable and submit M28750/10-002 (2 units) to group A inspection of MIL-R-28750.

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CONCLUDING MATERIAL

Ⓑ Custodians.
Navy - EC
Air Force - 85

Preparing activity:
Navy - EC

Agent:
DLA - EC

(Project 5945-0778-02)

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